

Task Description

There are () that make up proteins, each with unique chemical properties influencing the protein's structure and function. Proteins can vary in size, containing anywhere from a . A typical protein sequence can be represented as a string of single-letter codes for each residue, such as:

Example protein sequence: "MKTFFVFTTQRYDEQELFFQN" (M, K, T, ... are the protein's residues)

refer to pairs of residues that are spatially close within the 3D structure of the protein. In this task, () Identifying these contacts is crucial for understanding protein folding and interactions, as they provide insights into the overall structure and function of the molecule.

You are tasked with developing a machine learning model () acts within protein sequences. This model will be implemented in Python and trained using an open-source protein database that provides sequence and 3D structural data. The goal is to ()

() without explicitly predicting the full 3D configuration (i.e. atom coordinates).

In your solution you may ().

This model should build upon the () which already provides a () but

() The main goal of this task is to r () , enhancing the prediction performance.

Task Details

Objective: Create a new residue contact prediction model that leverages both the ESM2 embeddings of the input sequence residues and structural data from similar sequences.

Implementation:

Implement your model in Python, modifying or extending the ESM2 model to incorporate additional structural information. The input of the model is a single sequence, the output is a binary contact map (1 - if there is a contact between the residue pair, 0 - if there is no contact).

For the () data in () format. You may download the data from [here](#). When mapping protein residue 3-letter codes to their 1-letter code, you may use the table provided at the end of this document.

Deliverables:

- Submit your code as a ZIP file or upload it to a public GitHub repository.
- Include a concise report that outlines:
 - A precise description of your method and how it extends ESM2.
 - Data pre- and post-processing steps.
 - Hyperparameters and other relevant details of the ML setup.
 - Analysis of the results, including performance metrics and insights.

Evaluation Criteria

- Code quality
- ML training and evaluation setup, data handling
- Model performance
- Results analysis
- Method description

Protein residue codes mapping

- ALA: A
- ARG: R
- ASP: D
- CYS: C
- CYX: C
- GLN: Q
- GLU: E
- GLY: G
- HIS: H
- HIE: H
- ILE: I
- LEU: L
- LYS: K
- MET: M
- ASN: N
- PHE: F
- PRO: P
- SEC: U
- SER: S
- THR: T
- TRP: W
- TYR: Y
- VAL: V